INCH-POUND

MIL-S-24187/5A(SH)
22 March 1991

SUPERSEDING
MIL-S-24187/5(SH)
15 June 1978

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL OPERATION, STYLE JLS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

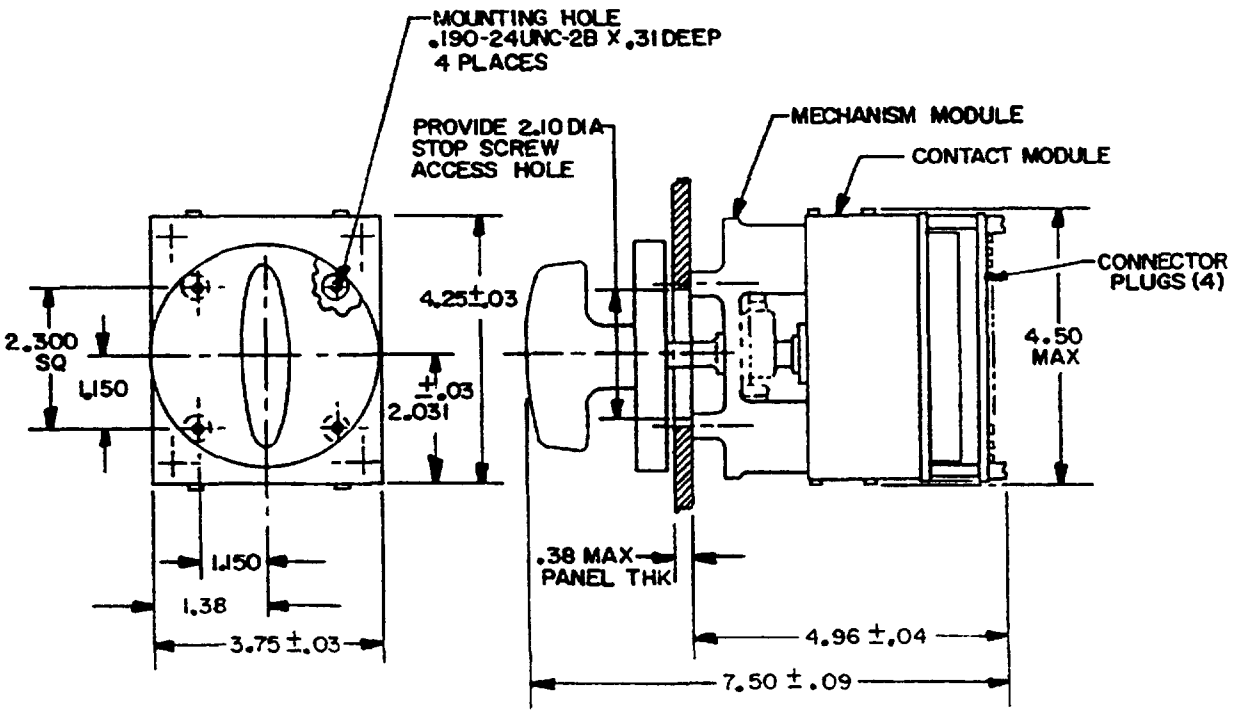


FIGURE 1. *JLS switch assembly*

AMSC N/A FSC 5930

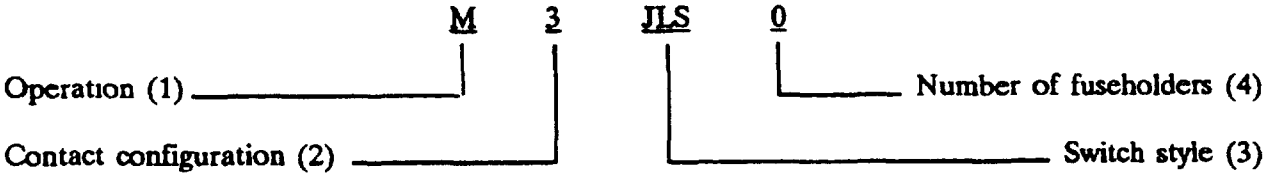
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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 for two-place decimals and plus or minus 0.010 for three-place decimals.

FIGURE 1. *JLS switch assembly – Continued.*



(1) Switch assemblies shall be manual operated designated m.
(2) Contact configuration shall be type 3 (see figure 3).
(3) Switch style shall be designated by the letters JLS.
(4) Fuseholder mounting capability shall not be provided in style JLS switches.

TABLE I. *Switch selection.*

Type designation				Application data
				Switch function and handle position
Operation	Contact configuration	Switch style	Fuseholders	1, 2, 3, 4
M	3	JLS	0	ON A ON B ON C OFF

EXAMPLE Type designation for a manual switch assembly, type 3 with no fuseholder is M3JLS0.

[1]Reference position of a JLS switch is considered to be that in which the flat on the handle shaft is up The mark ∇ and the dial positions shown are used to designate the reference position

[2]Normal position of a JLS switch is the position which coincides with normal circuit conditions in any application The handle shall be adjustable to point upward in the normal position or to any specified position, in 45 degree increments from upward The handle and rotary description plate rotate

[3]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated

[4]See figures 2 and 3 for complete description of contact identification

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NOTES:

1. 1-OFF-2 function is obtained by wiring loads to terminals A and C, using B as the OFF position and locating the stops to limit travel to the A, B, and C positions.
2. OFF-1 or OFF-1-2 is obtained by wiring load(s) to terminals C or C and B and locating the stops to limit travel as required.

REQUIREMENTS:

Construction: Style JLS switch parts, modules, and assemblies shall be constructed in accordance with Drawing 803-4680151.

Characteristics: Characteristics shown in table II shall apply to the style JLS switch assemblies.

TABLE II. *Characteristics.*

Characteristics	Value
	Contact configuration
Number of poles	56
Number of positions	4
Break-before-make	
Electrical rating and endurance loading:	
4 amperes, 28 V, dc, resistive	50,000 Operations
4 amperes, 115 V, ac, resistive	50,000 Operations
25 microamperes, 1.5 V, dc, unity p.f	50,000 Operations
Contact resistance:	
At 4 amperes	0.03 ohm
At 25 microamperes	0.05 ohm
Torque:	
Shaft rotation	20 pound-inches
Stop strength	100 pound-inches
Weight	4 pounds
Temperature and humidity (for rotational torque and endurance tests):	
Minimum	Minus 20 °F (minus 29 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except variable frequency
Temperature rise	30 °C

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Switch assembly: modules

The M3JLS switch assembly shall consist of the following modules as detailed on Drawing 803-4680151:

Mechanism module
Switch contact module

Switch function:

Style JLS switch assemblies shall provide 56, 4-position poles each consisting of 5 stationary contacts X, A, B, C, and D and a movable contact arranged as shown on figure 2. A stop shall prevent transfer of the movable contact into position X. Additional stops (removable) shall be provided which, when inserted as specified in table I will limit movement to two or three consecutive positions. Stops shall be as detailed on Drawing 803-4680151 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate.

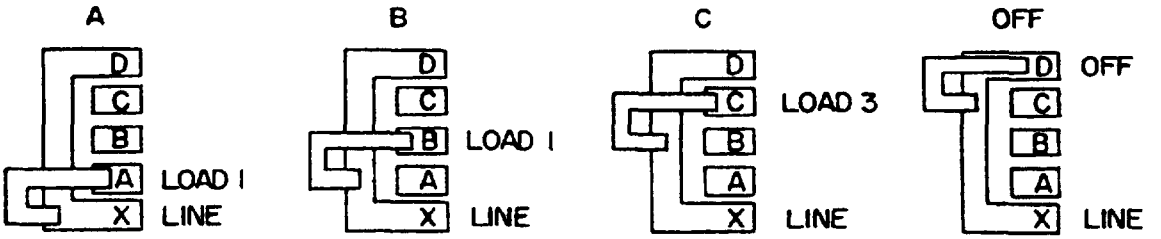


FIGURE 2. *Schematic typical operation of 1 pole.*

Contact designation:

Each stationary switching contact shall be identified by two letters and a number. The first letter, A, B, C, or D shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact D is common with contact X in all poles in the contact module; therefore, terminals for switching contacts D shall not be provided in the connector plugs.

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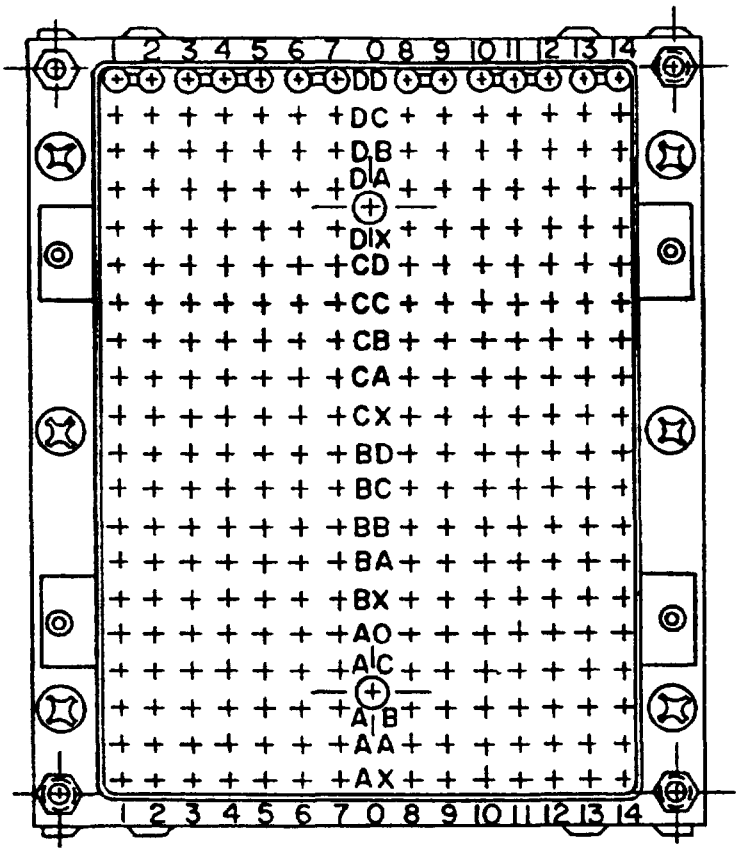


FIGURE 3. *Contact module marking – rear view.*

Repair parts: Repair parts shall be as specified in table III.

TABLE III. *JLS repair parts.*

Drawing 803-4680151 piece No.	Description	Quantities required for quantities of switches installed					
		1–3	4–12	13–30	31–60	61–120	121 or more
1	Switch assembly	1	1	1	1	2	2
3	Switch contact module	1	1	1	2	2	3
2	Mechanism module	1	1	1	2	2	3
43	Stop screw	1	1	1	2	2	3
16	Detent, spring	1	1	1	2	3	6
52	Washer indexing	See note					

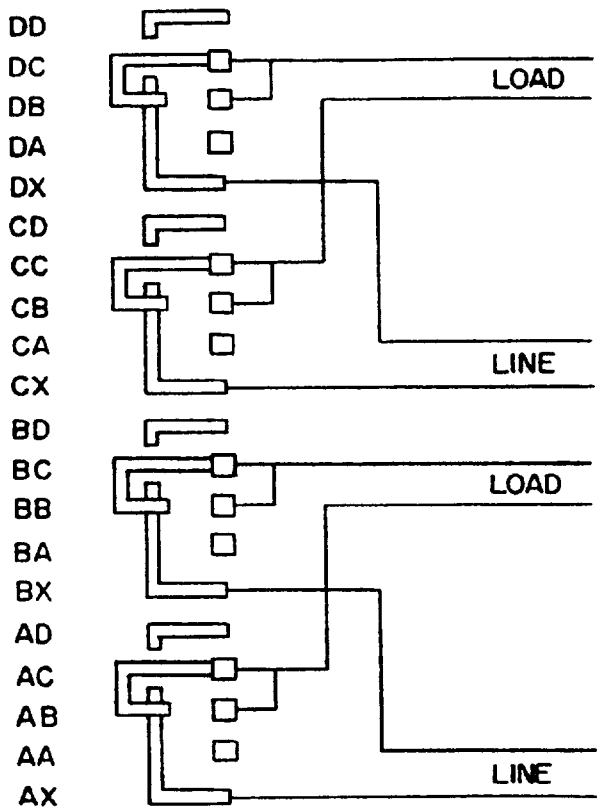
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NOTE: Supply one for each 10 or fraction of 10 installed.

Qualification samples: Two switch assemblies shall be submitted for qualification inspection.

TESTS:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be as shown on figure 4 and table IV.



NOTE: Rows 1, 5, 9, and 14 shall be loaded at 4 amperes resistive at 115 volts ac Rows 2, 6, 10, and 13 shall be loaded at 25 microamperes resistive at 1.5 volts dc Rows 3, 7, 8 and 12 shall be loaded at 4 amperes resistive at 28 volts dc.

FIGURE 4 *Test loading*

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Temperature rise: Contacts to be monitored during temperature rise tests are listed in table IV.

TABLE IV. *Temperature measurements.*

Column	Contact
1	AX, AB
1	CX, CD

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-06)

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